July 18, 1944.

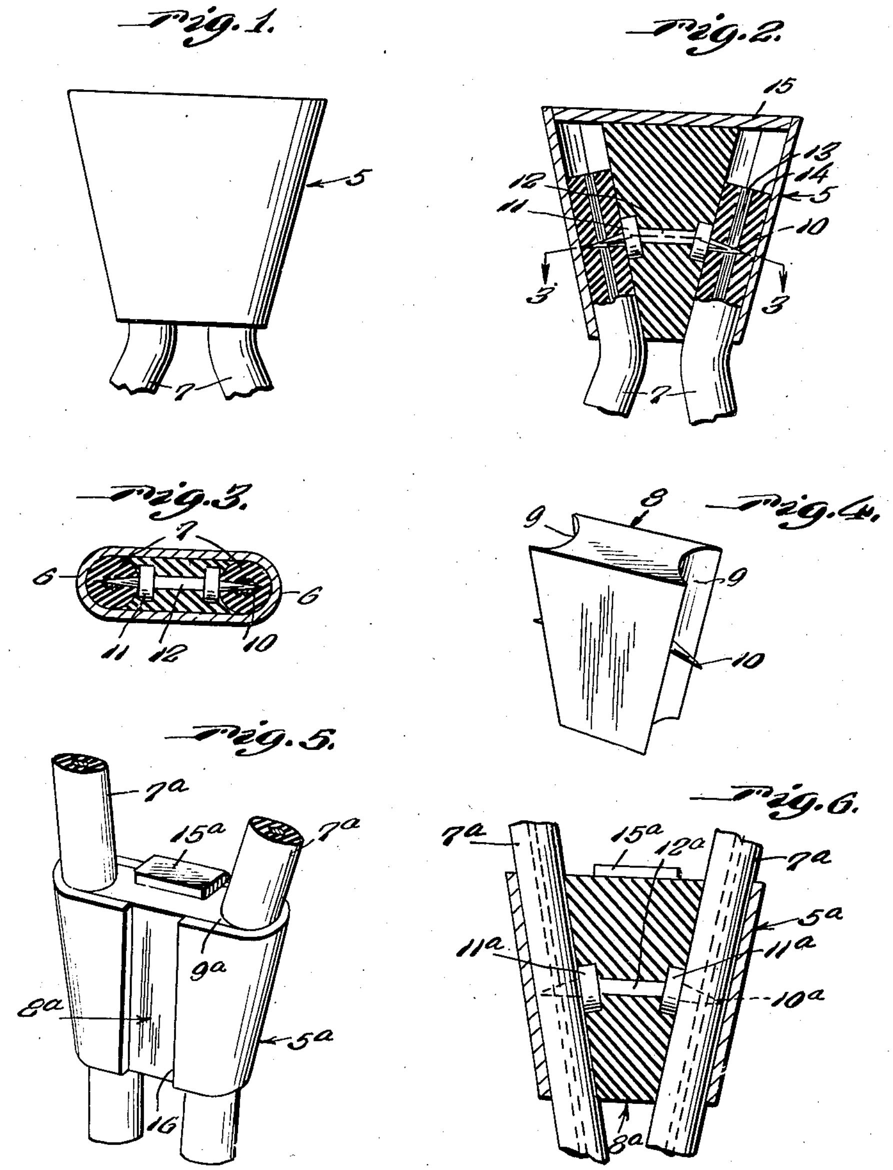
M. A. MATTIS

2,353,778

CONNECTOR FOR INSULATED CONDUCTORS

Filed Dec. 16, 1941

2 Sheets-Sheet 1



Inventor

Michael A. Mattis

By Clarence A. O'Brien

Attorney

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Fig. 7.

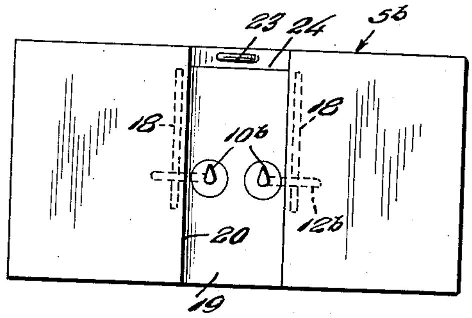


Fig. 9.

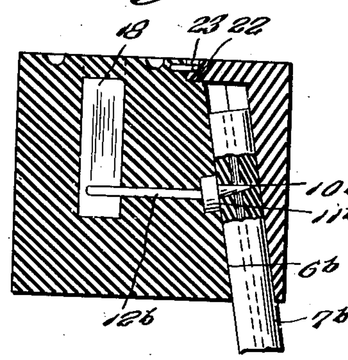


Fig. 8.

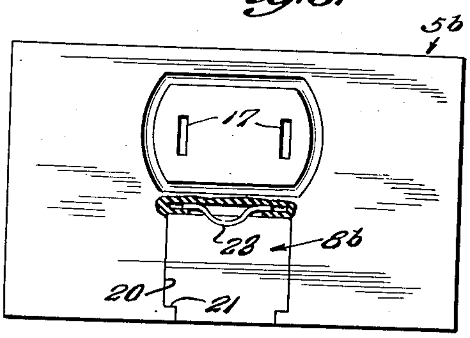


Fig. 10.

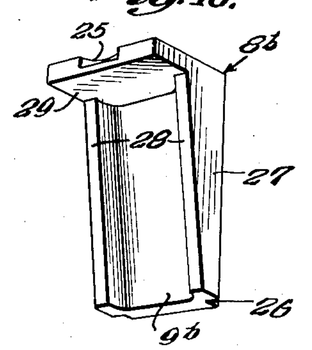


Fig. 11.

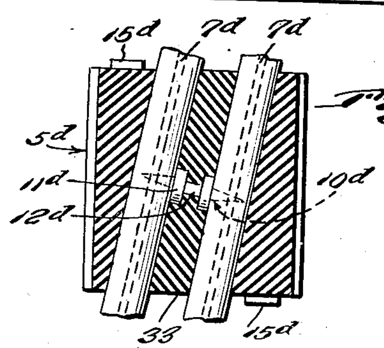


Fig. 13.

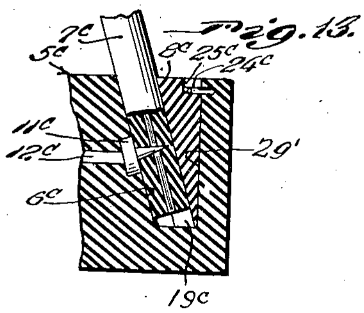
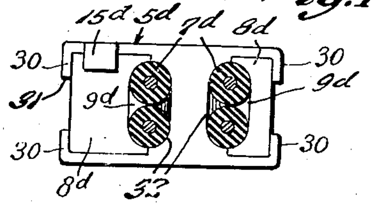


Fig. 12.



Inventor

Michael A. Mattis

By Clarence A. O'Brien

Attorney

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2,353,778

UNITED STATES PATENT OFFICE

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CONNECTOR FOR INSULATED CONDUCTORS

Michael A. Mattis, Shamokin, Pa.

Application December 16, 1941, Serial No. 423,268

4 Claims. (Cl. 173—349)

This invention relates to improvements in connectors for insulated conductors, and the primary object of the invention is to provide simple and efficient means operating on the wedge principle, whereby two or more insulated wires may be strongly and efficiently connected together both mechanically and electrically with a minimum amount of manipulation and handling.

Another important object of the invention is to provide improvements over the subject matter of my prior Patent No. 2,235,231 granted March 18, 1941, for Electrical connectors. The subject matter of the said patent was characterized by the entry of the wires into the connectors at the wide end of the wedge arrangement, the present invention offering increased efficiency and grip upon the wires by entering the wires into the connector from the narrow end of the wedge arrangement.

Other important objects and advantages of the invention will be apparent from a reading of the following description taken in connection with the appended drawings, wherein for purposes of illustration there is shown preferred embodiments of the invention.

In the drawings—

Figure 1 is a side elevational view of an embodiment of the invention for connecting terminal ends of two insulated wires to establish electrical connection therebetween.

Figure 2 is a vertical longitudinal sectional view taken through Figure 1.

Figure 3 is a horizontal sectional view taken through Figure 2 along the line 3—3 and looking downwardly in the direction of the arrows.

Figure 4 is a perspective view of the wedge employed therein.

Figure 5 is a perspective view of another embodiment of the invention for electrically connecting intermediate portions of two insulated wires.

Figure 6 is a vertical longitudinal sectional view taken through Figure 5.

Figure 7 is a front elevational view of an outlet characterized by the electrical connection of two insulated wires to the contacts thereof by removal wedge means.

Figure 8 is a top plan view of Figure 7 partly in section to disclose the spring detent for holding the wedge means.

Figure 9 is a transverse vertical sectional view taken through Figure 8.

Figure 10 is a rear perspective view of the wedge.

Figure 11 is a longitudinal vertical sectional view taken through a still further form of connector for electrically connecting intermediate portions of two insulated wires and wherein two wedge elements are employed.

Figure 12 is a top plan view of Figure 11.

Figure 13 is a fragmentary transverse vertical sectional view taken through a still further embodiment of the invention in an outlet similar to that shown in Figures 7 through 10 but employing a different wedge arrangement.

Referring in detail to the drawings, and first to Figures 1 through 4 thereof, the numeral 5 designates a truncated wedge-shaped casing of tubular form and made of metal or other suitable material, characterized by semi-circular ends 6 of approximately the same inside diameter as the insulated wires 7 which are to seat against the inner surfaces of said ends in the manner indicated in Figures 2 and 3 of the drawings with the wedge 8 forced between the wires.

The wedge 8 has substantial semi-circular grooves 9 in its opposite ends in which and beyond which project the impaling points 10 which project from and are preferably integral with circular metal bodies 11 seated in accommodating recesses in the bottoms of the grooves 9, and connected by a shank 12 extending within the wedge, as shown in Figure 2 of the drawings. The impaling points 10, the disks 11 and the connecting shank 12 are of suitable non-ferrous metal, and the points 10 are sufficiently long to project past the center of the wires 7 so as to project through the conductors 13 from one side of the insulation 14 to the other as indicated in Figures 2 and 3 of the drawings, and thereby make a good electrical contact with the conductors 13, particularly where the conductors are of the multiple flexible type. In applying the connector to the wires 7 the terminals of the wires are placed in the grooves 9 of the wedge 8 after having been passed through the lower end of the casing 5 and sufficient inward pressure is exerted to force the impaling points 10 through the conductors. The wedge and the conductors are then passed downwardly into the casing 5 from the larger end thereof so as to gradually compress the terminal ends of the wires 7 against the curved ends 6 of the casing between these ends and the grooves 9 of the wedge, until the wedge is finally forced home and the cover plate 15 seats into the upper or larger end of the casing 5 for which it is fitted, as indicated in Figure 2 of the drawings, whereupon the operation of forming the connection is complete. It will be observed that the wires so connected, when pulled in different directions wedge more securely in the connector, instead of having a tendency to separate or loosen therefrom.

Referring now to the form of the invention shown in Figures 5 and 6 of the drawings, this is closely related to the first described embodiment in that a truncated wedge-shaped tubular casing **5a** is present together with a similar wedge **8a** having the impaling points **10a**, the disks or buttons **11a** embedded in the bottoms of the grooves **9a** in the ends of the wedge and the connecting shank **12a** between the buttons **11a**. However, the present embodiment is intended for connecting intermediate portions of the insulated wires **7a** instead of the terminal portions thereof, and the wires pass entirely through the casing **5a** instead of terminating therein. Also, instead of the cover **15** to hold the wedge **8** in wedging position a similarly functioning element **15a** is provided in the form of a bent over ear integral with the upper edge of the casing **5** and which overlies the upper or wider end of the wedge as indicated in Figures 5 and 6. The casing **5a** differs further from the casing **5** by the presence of a gap or slot **16** in the side opposite that bearing the ear **15a**, whereas the casing in the first described embodiment has a continuous wall.

Referring to the embodiment of the invention shown in Figures 7 through 10, showing application of the principles of the present invention to a wall or base outlet or the like, the numeral **5b** generally designates a substantially rectangular block of insulating material having formed therein longitudinally spaced vertical slots **17** through which the prongs or blades of a plug (not shown) are to be inserted to make electrical contacts with the vertically elongate plate contacts **18** which are seated in the lower parts of the slots **17**. Secured in electrical connection with the contacts **18** are rods or shanks **12b** which project forwardly to connect with disks **11b** seated flush in the declining surface **6b** corresponding in function to the interior walls of the ends of the casings of the preceding embodiments of the invention, said disks having impaling points **10b** projecting outwardly from the surface **6b** as indicated in Figure 9 of the drawings to pass through the insulation and through or in contact with the conductors of the wires **7b**.

The declining surface **6b** is the back wall of a recess **19** having substantially parallel side walls **20** which is formed in the front of the block **5b** with the front portions of the side walls having inturned flanges **21**. The upper end of the wall **6b** is cut in as indicated by the numeral **22** to provide a ledge above which there projects from the block a bowed spring detent **23** as indicated in Figures 8 and 9 to engage in a notch **25** formed in the top of the wedge **8b** when the wedge is pressed downwardly into the recess **19** from the wide upper end thereof and with the grooves **26** in the outer corners of the wedge receiving the retaining flanges **21**, whereby the wedge is locked in applied position.

The wedge **8b** is characterized by the triangular cross section body portion **27** in whose outer corners the grooves **26** are formed and from whose outer corners project inwardly the inverted triangular side walls **28** which with the inner face of the body **27** define a relatively wide wedge groove corresponding in function to the grooves **9** and **9a** in the priorly described embodiments of the invention. The upper end of the body **27** is extended inwardly in a plate form as indicated by the numeral **29** to extend beyond the side walls **28**, the notch **25** being formed in the inner extremity of the plate **29**.

While the present embodiment is shown equipped to clamp and separately connect two insulated wires to distinct electrodes or contacts **18**, it is obvious that with respect to the present embodiment and all other embodiments herein disclosed, the invention is equally applicable to any practical number of wires. The present embodiment being particularly adapted to the connection of terminal ends of the wires **17**, the said terminals are put in place on the impaling points **10b** with the wedge **8b** removed, and then the wedge is forced down into place in the recess **19** until the detent **24** snaps into place into the notch **25** to hold the wedge in place and keep the wires impaled and clamped to the body **5b**.

In the embodiment of the invention shown in Figure 13 of the drawings, the same is similar to that shown in Figures 7 through 10 except that the recess **19c** does not open through the front of the body **6c** and the wedge **8c** is of a solid inverted type which works against the side of the wire **7c** in opposition to the declining surface **6c** and against the right angular surface **29'** which is opposed to and spaced from the declining surface **6c**, all as shown in Figure 13 of the drawings. A spring detent **24c** is located at the upper end of the right angular wall **29** to engage in the notch **25c** in the upper end of the wedge **8c** to hold the wedge removably in place.

Referring now to the embodiment of the invention shown in Figures 11 and 12 of the drawings, this is especially adapted for making electrical connection between intermediate portions of several insulated wires, with the wires arranged in substantial parallel relationship. This particular embodiment is characterized by a casing **5d** which is of generally rectangular cross section and suitable insulation material and provides abbreviated end walls or flanges **30** which are spaced from each other as indicated by the numeral **31** and against which the unangulated backs of the wedges **8d** slide, with the grooved inclined and declined surfaces **9d** of the wedges engaged with the sides of the wires **7d**, with the wedges reversed as indicated in Figure 11 of the drawings and confined between the side walls of the casing as indicated in Figure 12. The inner sides of the wires **7** bear against the surfaces of grooves **32** formed in an insulated intermediate partition forming body **33** integral with the casing **5d** and in which are embedded the conductive disks **11d** connected by the conductive shank **12d**, the disks being flush with the grooves **32** and provided with impaling pionts **10d** passing through the wires and making contact with the conductors therein. It will be obvious that the opposed wedges **8d** work against each other in forcing the wires against the opposite sides of the intermediate insulated body **33** and that it is necessary to apply the wires to the opposite sides of the intermediate body after passing the wires and the body through the casing before placing the wedges and forcing the same into clamping relationship. After the wedges are in operative position tongues or ears **15d** formed on the side walls of the casing are bent over the wide ends of the wedges to hold the same in place.

Although there is shown and described herein preferred embodiments of the invention, it is to be definitely understood that it is not wished to limit the application to the invention thereto except as may be requred by the scope of the subjoined claims.

Having described the invention, what is claimed as new is:

1. An electrical connector for at least two insulated wires, said connector comprising a casing formed with a tapering recess affording two opposed wedging surfaces and into the small end of which the wires are adapted to be inserted in the recess to lie along the said wedging surfaces, and a wedge inserted through the wider end of said recess between the wires lying along said wedging surfaces, said wedge having its working faces forcibly engaging the wires, conductive impaling points projecting from said working faces and piercing the insulation and engaging the conductors of said wires.

2. An electrical connector for at least two insulated wires, said connector comprising a casing formed with a tapering recess affording two opposed wedging surfaces and into the small end of which the wires are adapted to be inserted in the recess to lie along the said wedging surfaces, and a wedge inserted through the wider end of said recess between the wires lying along said wedging surfaces, said wedge having its working faces forcibly engaging the wires, conductive impaling points projecting from said working faces and piercing the insulation and engaging the conductors of said wires, and conductive elements connecting some of said impaling points together to electrically pair some of said wires.

3. An electrical connector for at least two insulated wires, said connector comprising a casing formed with a tapering recess affording two opposed wedging surfaces and into the small end of which the wires are adapted to be inserted in the recess to lie along the said wedging surfaces, and a wedge inserted through the wider end of said recess between the wires lying along said wedging surfaces, said wedge having its working faces forcibly engaging the wires, conductive impaling points projecting from said working faces and piercing the insulation and engaging the conductors of said wires, said wires passing out through the wider end of said recess.

4. An electrical connector for at least two insulated wires, said connector comprising a casing formed with a tapering recess affording two opposed wedging surfaces and into the small end of which the wires are adapted to be inserted in the recess to lie along the said wedging surfaces, and a wedge inserted through the wider end of said recess between the wires lying along said wedging surfaces, said wedge having its working faces forcibly engaging the wires, conductive impaling points projecting from said working faces and piercing the insulation and engaging the conductors of said wires, said wires terminating within said recess.

MICHAEL A. MATTIS.